FaceDoor — Quick Start Guide

CSIS 4495 Applied Research Project · Douglas College · Winter 2026

Requirements

Tool	Version
Python	3.11
Node.js	18 LTS or newer
npm	comes with Node.js
Webcam	required for face capture & detection

> **Windows only:** Run `pip install cmake` before step 1 if `face_recognition` fails to install.

Page 1 — Installation (One Time Only)

Step 1 — Install Python dependencies

Open a terminal, navigate into the `Implementation/` folder, then run:

```
# Mac / Linux
cd Implementation
pip install -r requirements.txt

# Windows PowerShell
cd Implementation
pip install -r requirements.txt
```

Step 2 — Install frontend dependencies

```
cd frontend
npm install
cd ..
```

> This also automatically downloads the MediaPipe pose model needed for fall detection.

Step 3 — (Optional) Register faces for real recognition

Skip this if you just want to see the dashboard with demo data.

```
# Step 3a — Capture photos from your webcam
python scripts/capture_faces.py

# Step 3b — Load those photos into the system
python scripts/register_faces.py
```

Page 2 — Running the Project

You need **two terminals open at the same time.**

Terminal 1 — Start the Backend (Flask API)

```
# Mac / Linux
cd Implementation
FLASK_PORT=5001 python main.py

# Windows PowerShell
cd Implementation
$env:FLASK_PORT=5001; python main.py
```

You should see:

```
Flask API running on http://0.0.0.0:5001
```

Terminal 2 — Start the Frontend (Next.js)

```
cd Implementation/frontend
npm run dev
```

You should see:

```
▲ Next.js ready on http://localhost:3000
```

Open the Dashboard

Go to **http://localhost:3000** in your browser.

Page	URL
Main Dashboard	http://localhost:3000
Alerts	http://localhost:3000/alerts
Access Logs	http://localhost:3000/logs
Audit Trail	http://localhost:3000/compliance
Fall Detection	http://localhost:3000/falls
Demo Center	http://localhost:3000/demo

> If no faces are registered the dashboard shows realistic **demo data** automatically — no setup required to explore the UI.

Optional — Enable Live Fall Detection (Phase 2)

Open a **third terminal** while the backend and frontend are already running:

```
cd Implementation
python scripts/fall_detection_camera.py --lstm
```

Then open **http://localhost:3000/falls** to see live LSTM confidence scores.

Press `Q` in the camera window to quit, or `R` to reset fall history.

Stopping the Project

- Press **Ctrl + C** in each terminal to stop the backend and frontend.
- The database keeps all logged events — just restart the two terminals next time.

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